



TEXAS A&M
UNIVERSITY

Relating Aversive Conditioning and Emotional Valence in the Control of Attention

Justin Frandsen, Francesca Canturi, Vera Ferrari, and Brian A. Anderson

Attention &
Learning
Lab

Introduction

- Images with high emotional valence (e.g., images of gore or violence) can capture attention.
- Stimuli previously associated with aversive outcome (e.g., previously associated with shock) can capture attention.
- Whether emotional content and aversive conditioning influence attention via common or dissociable mechanisms of attentional control remains unclear.

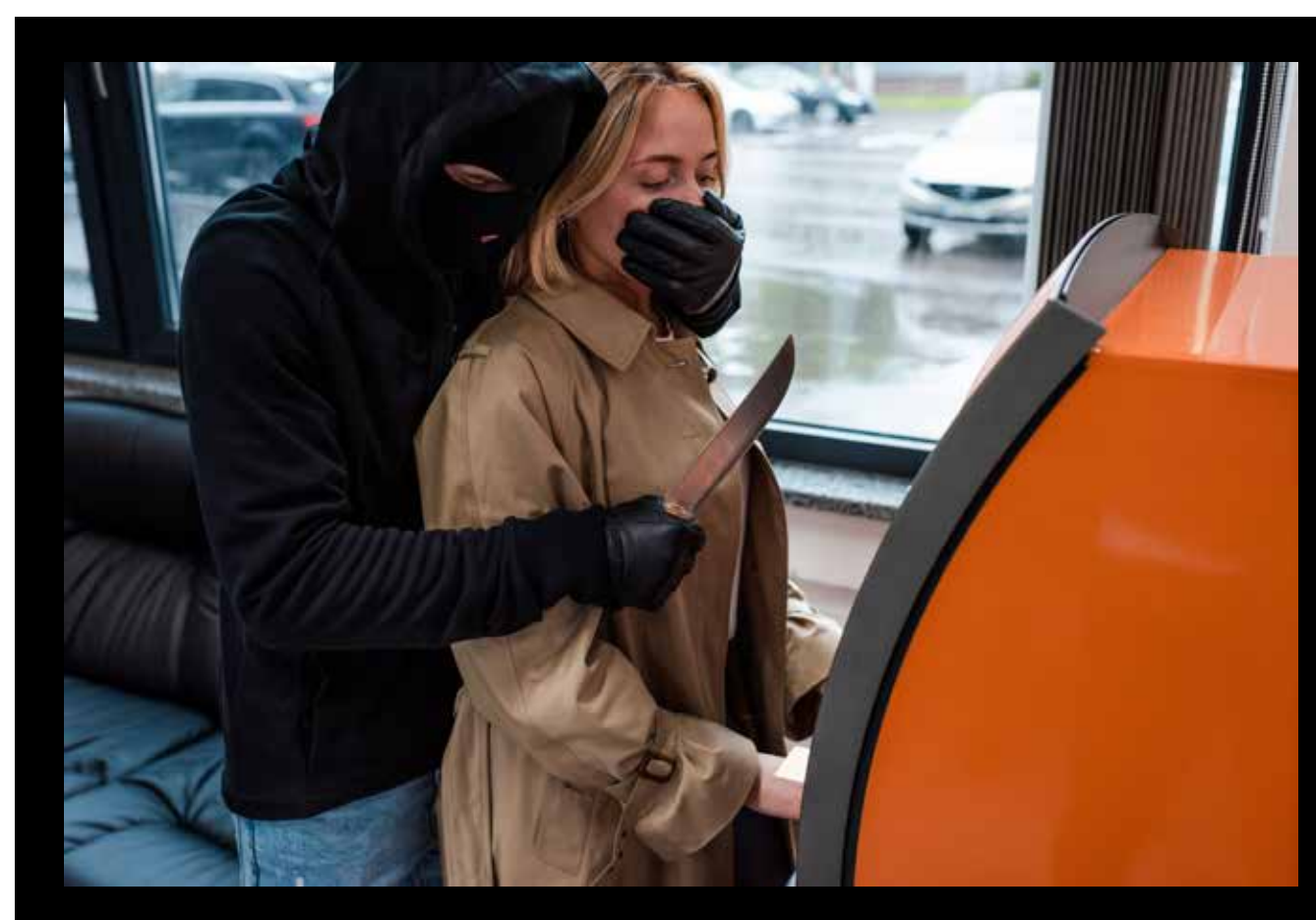
Methods

Training Phase

Shock Associated

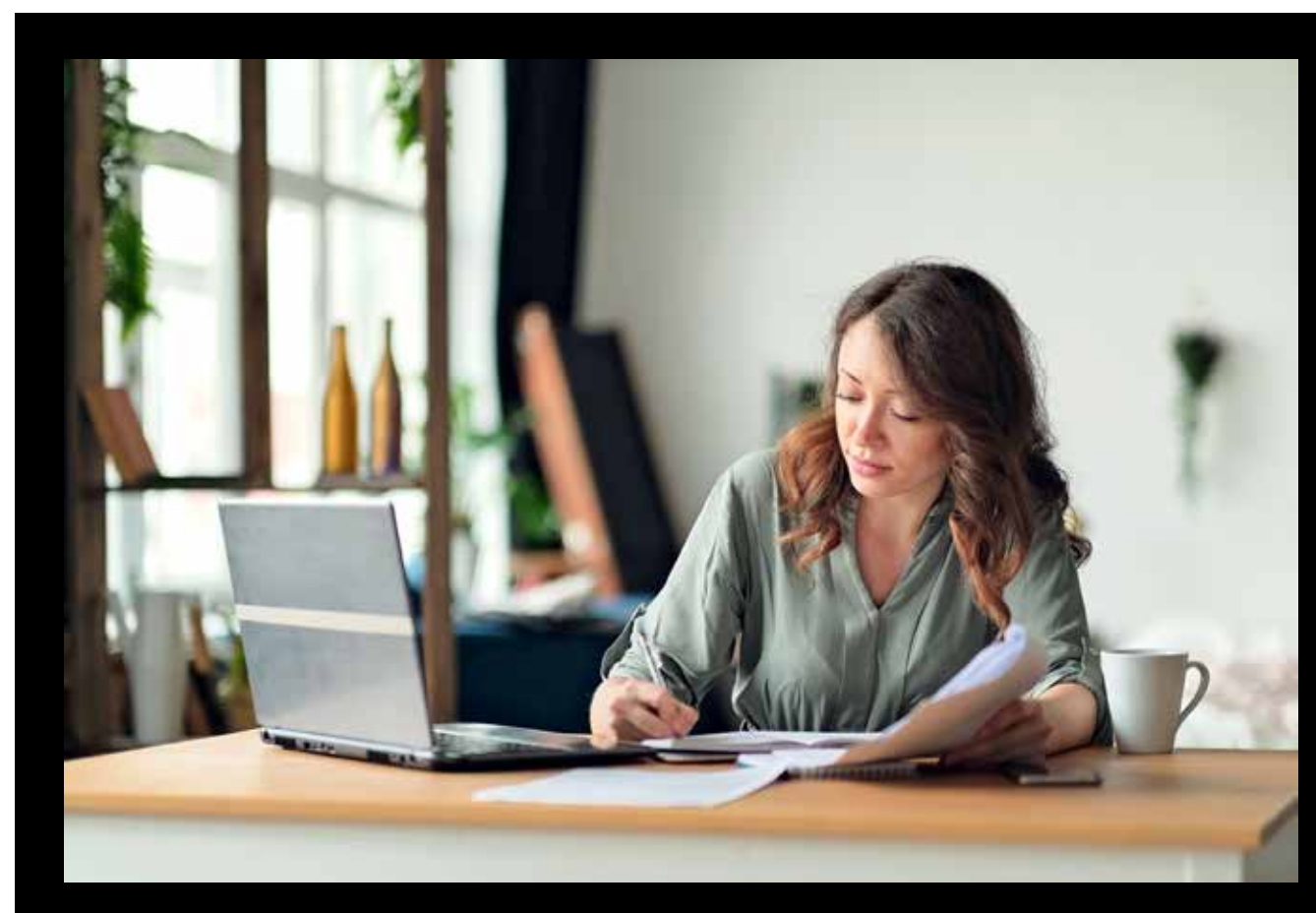


Non-shock Associated



- 10 blocks:
- 4 new images each block
 - 3 presentations per image

Neutral



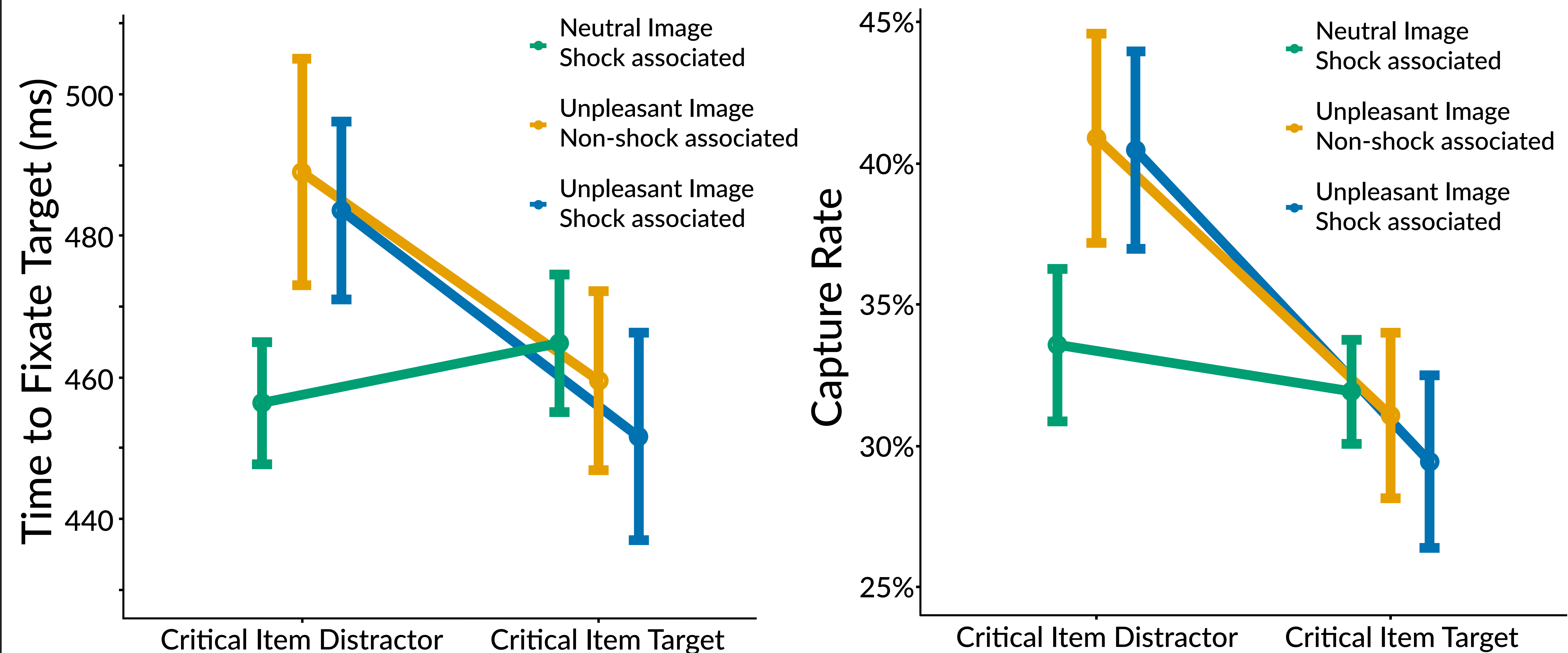
Testing Phase



+



Results



Critical Item (Target or Distractor)



+

Neutral Item



Conclusion

- Unpleasant images capture attention.
- Shock associated images did not capture attention under the conditions tested.
- The experience did not change the effect of capture by unpleasant images.

- Bradley, M. & Lang, P. (2007). The International Affective Picture System (IAPS) in the Study of Emotion and Attention
- Lundqvist, D., Esteves, F., & Ohman, A. (1999). Cognition and Emotion
- Mulckhuyse, M. (2018). Cognitive, Affective, & Behavioral Neuroscience
- Vuilleumier, P. (2002). Current Opinion in Psychiatry.



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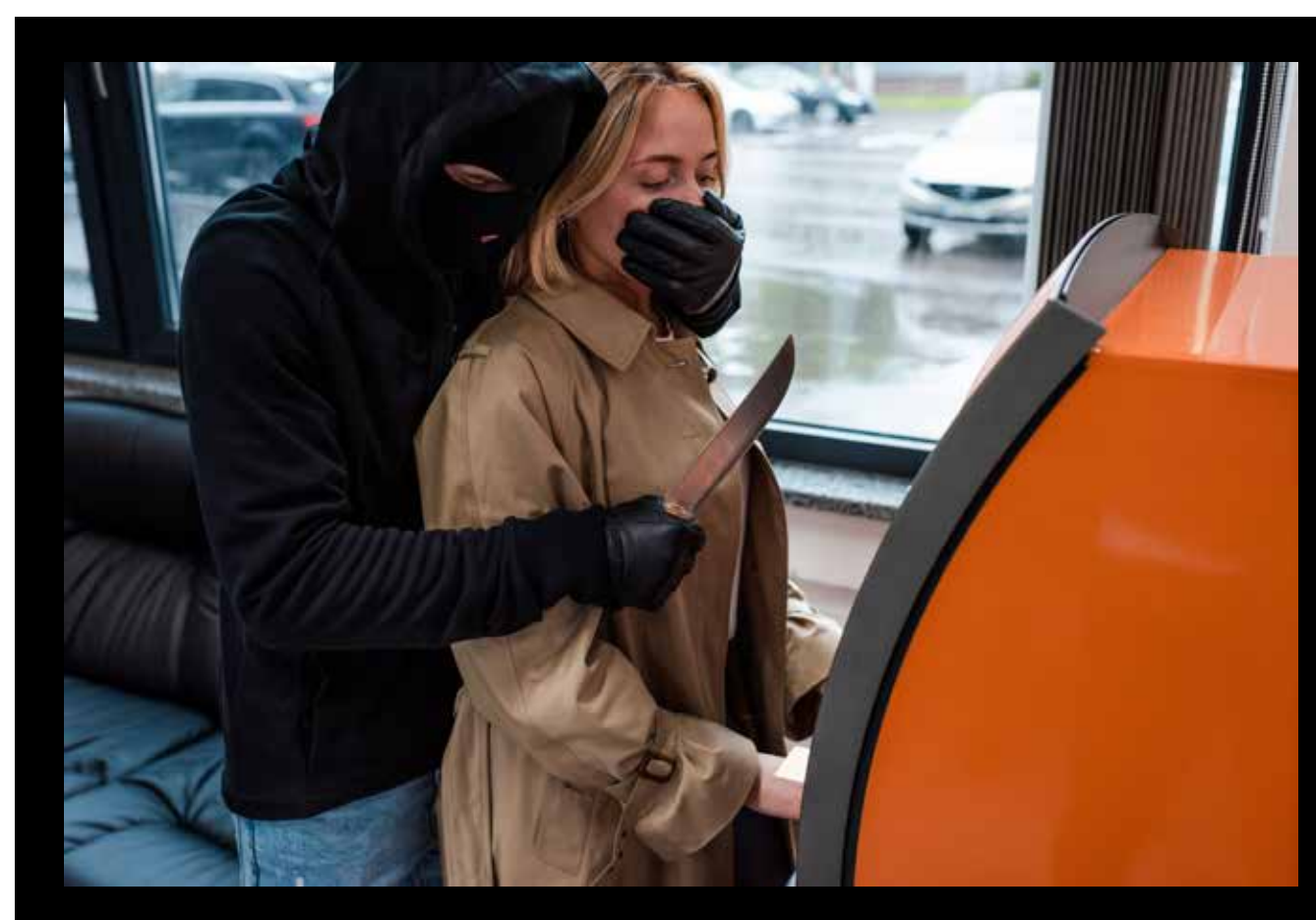
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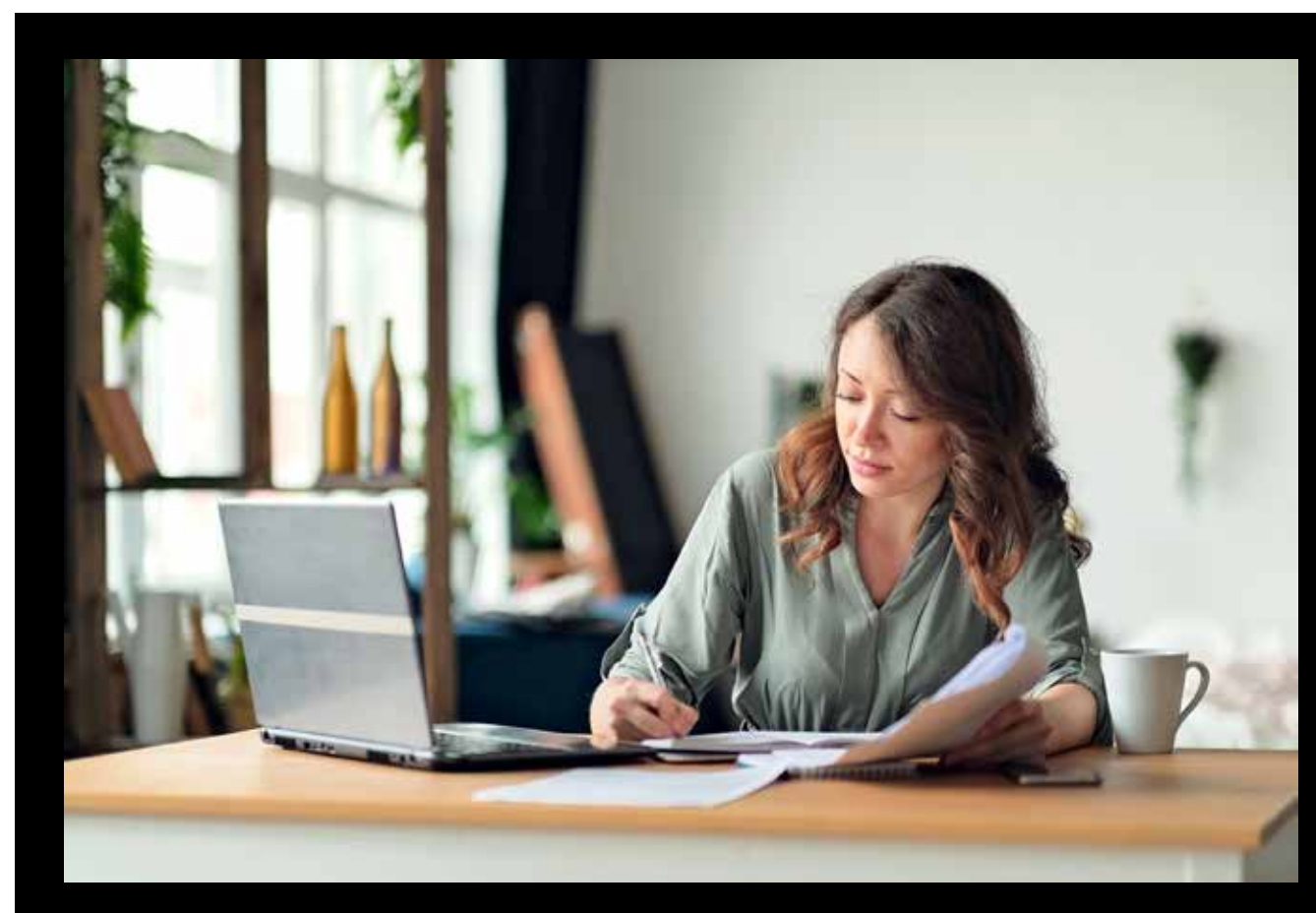


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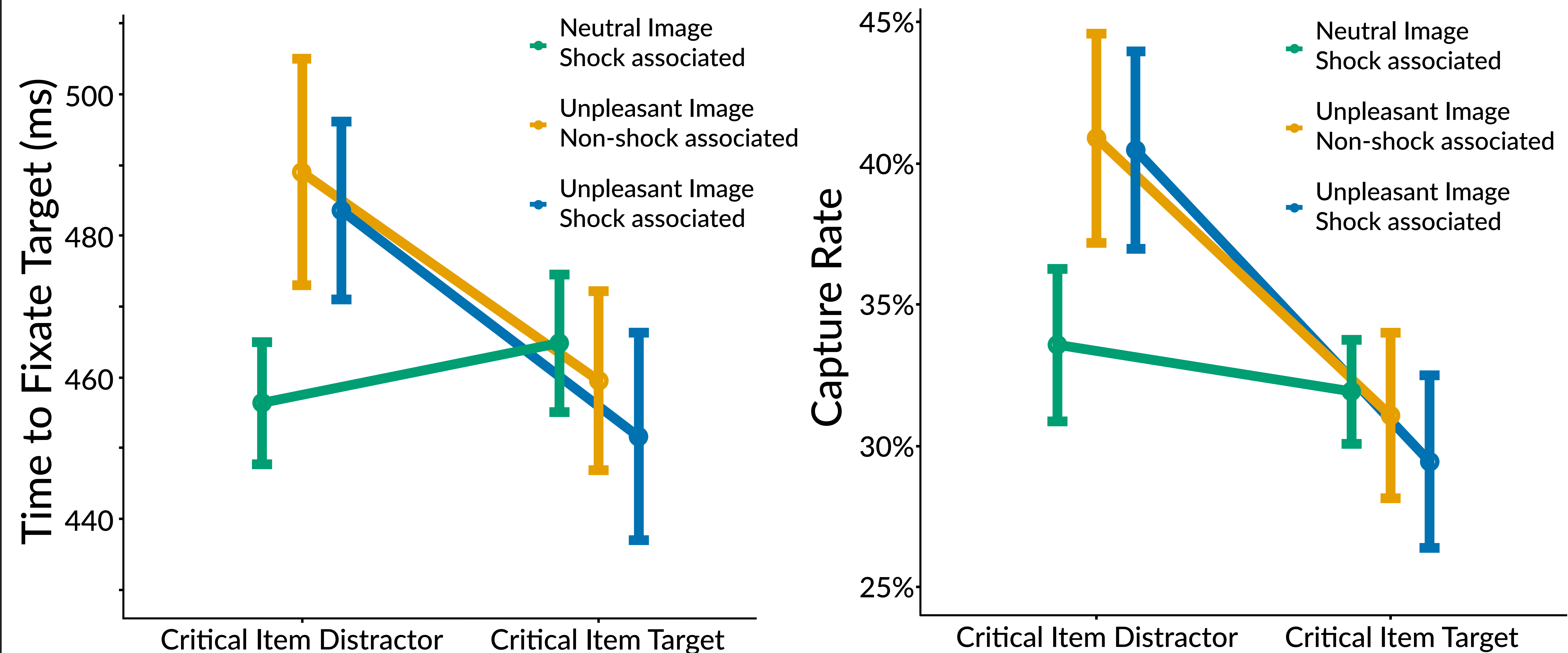
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